SOUTH KOREA RENEWABLE ENERGY

Economic and Environmental Analysis | 2015–2024

An Analysis of Electricity Generation,
Carbon Intensity, and Economic Growth

RESEARCH PERIOD	2015–2024
GEOGRAPHIC FOCUS	South Korea
ANALYTICAL APPROACH	Python and statistical analysis
RESEARCH AREA	Energy and environmental economics

Independent Undergraduate Research Project

Abstract

This study examines the relationship between renewable electricity adoption, electricity generation composition, carbon intensity, and economic growth in South Korea from 2015 through 2024. Publicly available economic and energy datasets were combined and analyzed using Python, descriptive statistics, correlation analysis, and data visualization.

Renewable electricity share increased from 2.77% in 2015 to 9.71% in 2024. Coal's share of electricity generation declined, while natural gas and renewable generation increased. Renewable electricity share was strongly negatively correlated with electricity carbon intensity (Pearson r = -0.8819, p = 0.0007). Real GDP increased by approximately 24.56%, while electricity carbon intensity declined by approximately 14.79%.

These findings indicate an association between changes in South Korea's electricity generation mix and declining electricity carbon intensity. They do not establish causation.

1. Introduction

The transition toward renewable energy presents an important economic and environmental challenge. Governments must balance energy security, economic growth, electricity demand, and emissions reduction while changing their energy systems.

This project investigates how changes in renewable electricity adoption in South Korea relate to electricity generation patterns, electricity carbon intensity, and economic growth.

Research Question

To what extent has renewable energy adoption in South Korea been associated with changes in electricity generation, carbon intensity, and economic growth?

Research Objectives

1. Analyze renewable electricity adoption from 2015 to 2024.
2. Examine changes in the electricity generation mix.
3. Investigate the relationship between renewable electricity and electricity carbon intensity.
4. Evaluate relationships between economic growth and annual changes in carbon intensity.
5. Identify potential economic and environmental implications.

2. Data and Methodology

Data Sources

Economic indicators were obtained from the World Bank's World Development Indicators: GDP in constant 2015 US dollars (NY.GDP.MKTP.KD) and annual GDP growth (NY.GDP.MKTP.KD.ZG). Electricity generation and carbon intensity indicators were obtained from Our World in Data and its underlying energy data sources.

The electricity dataset includes renewable electricity share, electricity carbon intensity in grams of CO2-equivalent per kilowatt-hour, and generation shares for coal, natural gas, nuclear, and renewables. A residual category was calculated as other sources.

Analytical Methods

The final dataset contains 10 annual observations from 2015 through 2024. Python libraries included Pandas, NumPy, Matplotlib, Seaborn, SciPy, and Requests. The analysis used descriptive trend analysis, Pearson correlation, Spearman rank correlation, and a sensitivity analysis excluding 2020 from one GDP-growth comparison.

A 5% significance threshold was used. Correlation coefficients describe associations and do not establish causal relationships.

3. Results

3.1 Renewable Electricity Adoption

Renewable electricity share increased from 2.77% in 2015 to 9.71% in 2024, an increase of approximately 6.93 percentage points. Renewables remained a relatively small share of total electricity generation in 2024 despite substantial growth.

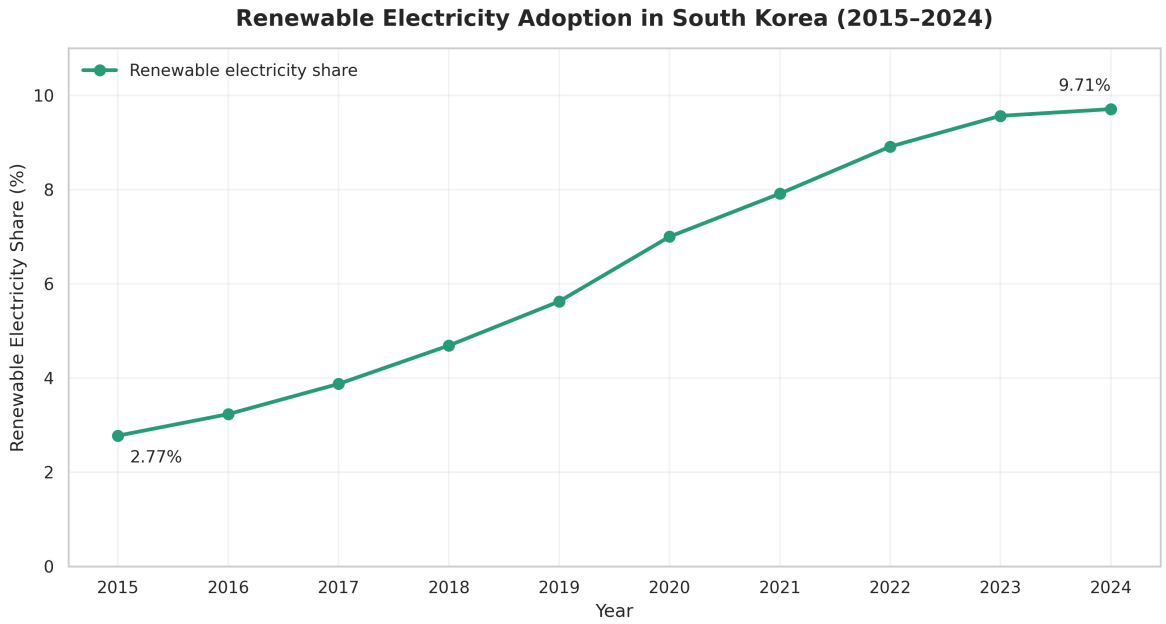


Figure 1. Renewable electricity adoption in South Korea, 2015–2024.

3.2 Electricity Generation Mix

Between 2015 and 2024, coal declined from 42.48% to 30.51% of electricity generation. Natural gas increased from 22.28% to 28.52%, nuclear changed from 30.08% to 30.18%, and renewables increased from 2.77% to 9.71%. These changes indicate a shifting generation composition rather than a transition driven by one source alone.

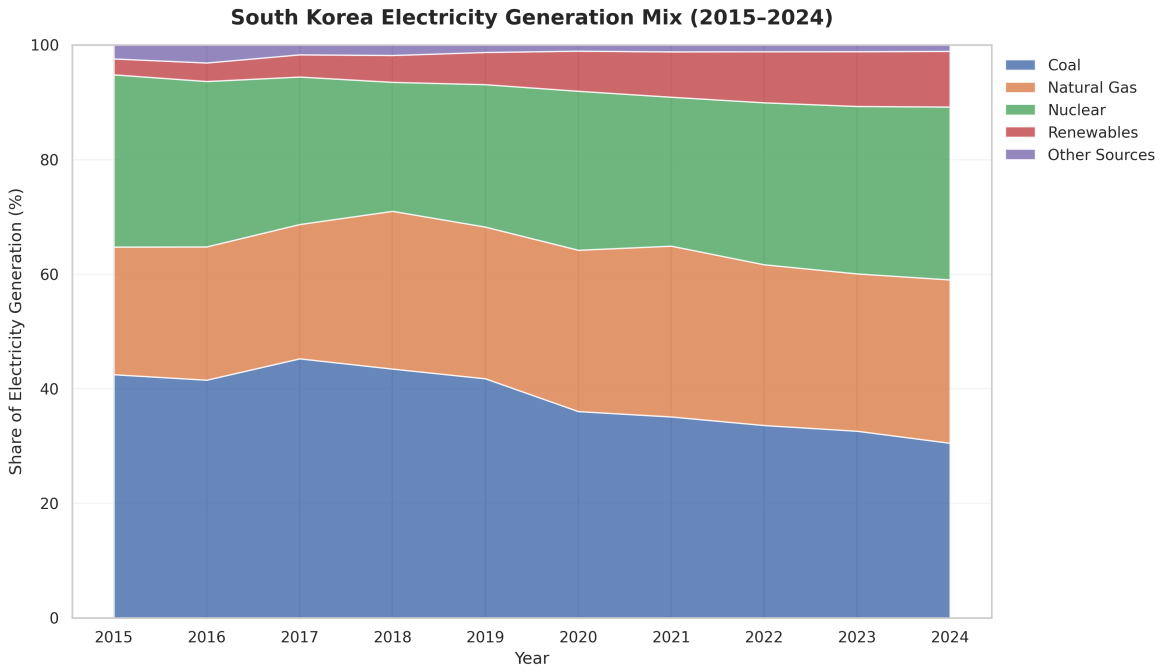


Figure 2. Electricity generation mix in South Korea, 2015–2024.

3.3 Renewable Electricity and Carbon Intensity

Renewable electricity share had a strong negative association with electricity carbon intensity. Pearson correlation was -0.8819 (p = 0.0007), while Spearman correlation was -0.7818 (p = 0.0075). Years with higher renewable shares tended to have lower electricity carbon intensity.

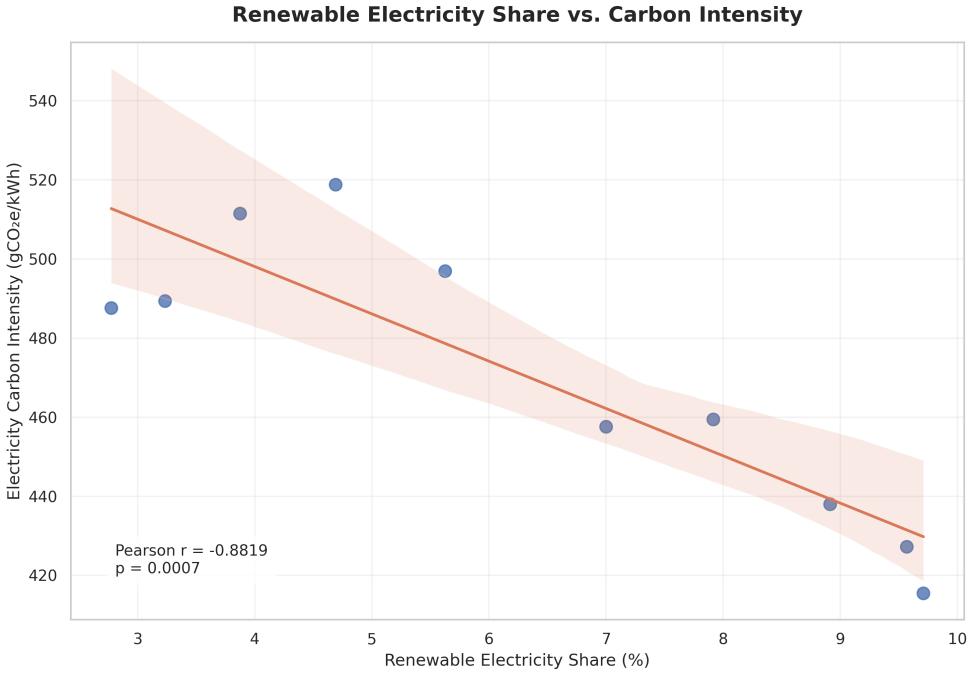


Figure 3. Renewable electricity share versus electricity carbon intensity.

3.4 Economic Growth and Carbon Intensity

From 2015 to 2024, real GDP increased by approximately 24.56%, while electricity carbon intensity declined by approximately 14.79%. This pattern is consistent with relative decoupling between economic growth and electricity-sector carbon intensity, not necessarily economy-wide emissions decoupling.

Annual GDP growth and annual changes in electricity carbon intensity had a positive association (Pearson r = 0.7775, p = 0.0137; Spearman r = 0.7833, p = 0.0125). Years with stronger GDP growth tended to coincide with less favorable annual changes in electricity carbon intensity.

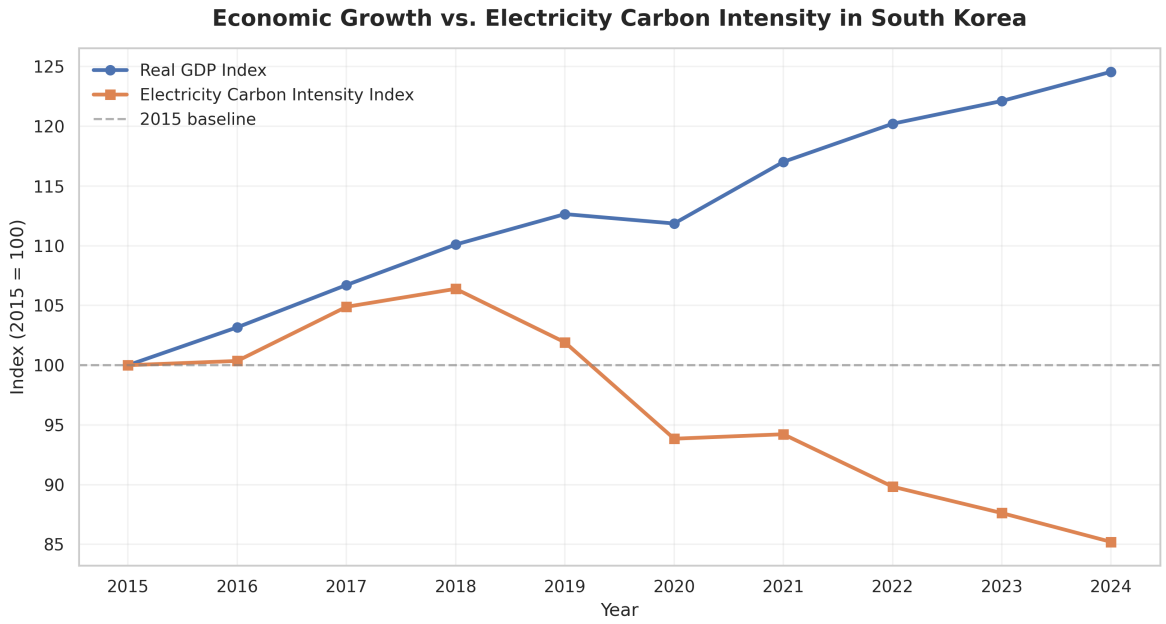


Figure 4. Real GDP and electricity carbon intensity indices (2015 = 100).

3.5 Electricity Sources and Carbon Intensity

Coal had a strong positive association with electricity carbon intensity, while nuclear and renewable electricity had negative associations. Natural gas also had a negative coefficient, but its relationship was not statistically significant at the 5% level.

Electricity source	Pearson r	p-value
Coal	+0.9781	<0.0001
Natural gas	-0.5642	0.0893
Nuclear	-0.6767	0.0316
Renewables	-0.8819	0.0007

Generation shares are interdependent because they collectively represent the electricity mix. These correlations should not be interpreted as independent effects.

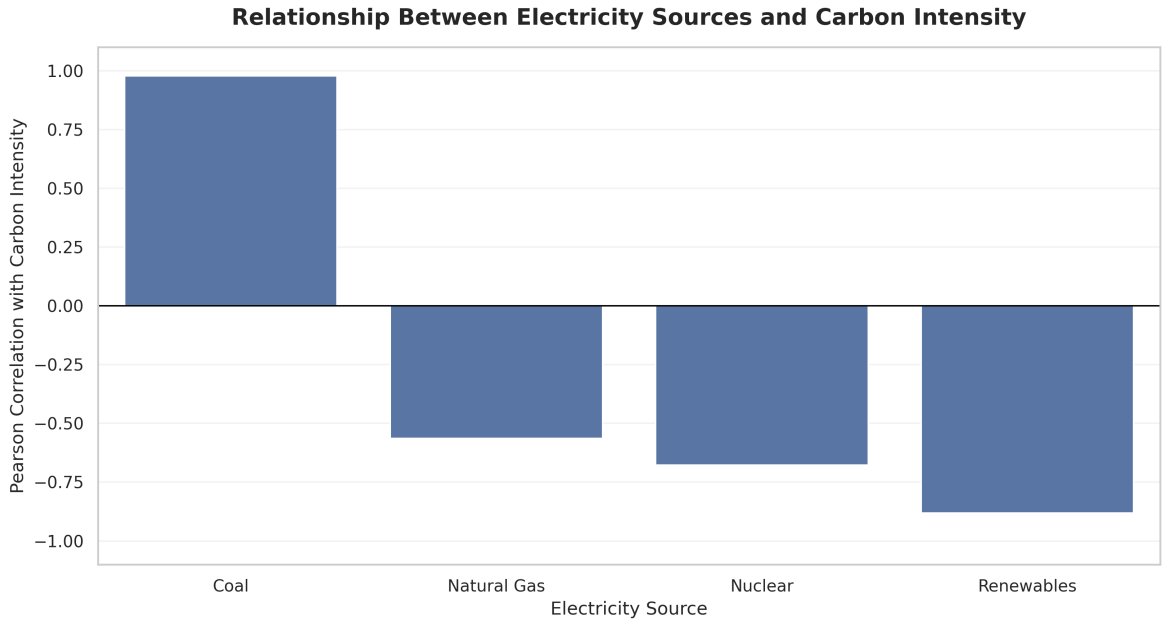


Figure 5. Pearson correlations between electricity generation shares and electricity carbon intensity.

4. Discussion

South Korea's electricity generation mix changed substantially between 2015 and 2024. Renewable electricity expanded, coal's generation share declined, and electricity carbon intensity decreased. The strong negative correlation between renewable share and carbon intensity is consistent with the possibility that renewable expansion contributed to a cleaner electricity mix.

Economic growth also continued over the study period. Real GDP increased while electricity carbon intensity declined, suggesting that economic expansion and improvements in electricity-sector carbon intensity can occur simultaneously. Other energy-system factors may also influence these results.

5. Limitations

The analysis contains only 10 annual observations, limiting statistical power. Correlations cannot establish causality. The carbon intensity measure applies to electricity generation rather than total national emissions. Generation shares are interdependent, and the analysis does not independently control for electricity demand, fuel prices, policy changes, or technological developments. Secondary datasets may also differ in measurement methods or revisions.

6. Conclusion

Between 2015 and 2024, South Korea experienced an expansion in renewable electricity generation alongside a decline in coal's share of the electricity mix and a reduction in electricity carbon intensity.

The analysis identified a strong negative association between renewable electricity share and electricity carbon intensity. Real GDP increased while electricity carbon intensity declined over the same period. These findings suggest that South Korea's changing electricity generation mix coincided with improvements in electricity-sector carbon intensity while the economy continued to grow. Further research is needed to identify the drivers.

7. Future Research

Future research could compare South Korea with other OECD economies, examine renewable energy investment and electricity prices, incorporate electricity demand and policy changes, and use longer time series and multivariable statistical methods.

8. References and Data Documentation

World Bank. World Development Indicators. GDP (constant 2015 US$), indicator NY.GDP.MKTP.KD; GDP growth (annual %), indicator NY.GDP.MKTP.KD.ZG.

Our World in Data. Energy and electricity generation datasets, including renewable electricity share, electricity generation mix, and electricity carbon intensity.

Before public release, add exact dataset URLs, access dates, and underlying source documentation from the notebook's data collection cells.

Project Tools

Python | Pandas | NumPy | Matplotlib | Seaborn | SciPy | Data Cleaning | Statistical Analysis | Data Visualization